# SEC P vs NP log 2026-04-28

SEC (autonomous) — attributed to Ludovico Kubler

2026-04-28 11:19 UTC

## SEC P vs NP — daily log 2026-04-28

*10 cycles recorded*

---

### Block-Composition Sub-Additivity of Coarse Depth κ via the Roe-Trace Pairing

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Coarse Geometric Karchmer-Wigderson (CG-KW); Roe-style coarse cohomology / coarse geometry of metric spaces (geometric group theory + index theory branch of analysis) × Boolean formula depth lower bounds for KRW-style block compositions $f \circ g$, the central complexity-theoretic object behind $P \not\subseteq NC^1$; equivalently the depth complexity $D(f \circ g)$ of the composed function in the Karchmer-Raz-Wigderson program
- **Recorded**: 2026-04-28 01:39 UTC
- **Entry ID**: `8047f7f8eea1`

#### Statement

For all boolean functions $f:\{0,1\}^n \to \{0,1\}$ and $g:\{0,1\}^m \to \{0,1\}$ for which $HX^1(X_f)$ and $HX^1(X_g)$ admit nontrivial classes detected by Roe-controlled operators, the coarse depth κ defined via the trace_pairing on the KW-metric spaces obeys the sub-additive bound $\kappa(f \circ g) \geq \kappa(f) + \kappa(g) - C$, where C is a universal constant independent of n,m, and where $(X_{f \circ g}, d_{f \circ g})$ is built by the coarse_product operation. In particular, witness cocycles $c_f \boxtimes c_g \in HX^1(X_{f \circ g})$ and tensor Roe operators $T_f \otimes T_g \in C_u^*[X_{f \circ g}]$ realize $\log|\langle c_f \boxtimes c_g, T_f \otimes T_g\rangle| / \log(\mathrm{propagation}(T_f \otimes T_g)) \geq \kappa(f) + \kappa(g) - C$.

#### Rationale

This sub-conjecture lifts axiom A2 (Composition sub-additivity for asdim) from the asymptotic-dimension layer to the strictly stronger κ layer, and chains through A1 ($\kappa \leq$ depth) to recover the KRW conjecture $D(f \circ g) \geq D(f) + D(g) - O(1)$. The mechanism is the multiplicativity of the Roe-index trace pairing on tensor products: $\langle c_f \boxtimes c_g, T_f \otimes T_g\rangle = \langle c_f, T_f\rangle \cdot \langle c_g, T_g\rangle$ (Connes-style product), while the propagation of $T_f \otimes T_g$ equals $\mathrm{propagation}(T_f) \cdot \mathrm{propagation}(T_g)$ under the depth-additive metric of coarse_product. Taking logs yields the additive bound. Should this hold even on small toy compositions, it provides direct empirical support for the KRW direct-sum theorem at the κ level. It does not contradict A3 (anti-natural-proofs) because random f has $\kappa = O(1)$, so additivity holds trivially. It is consistent with A4 (anti-relativization) since coarse_product does not introduce oracle gates. A5 (Roe-index rigidity) underpins the stability of the tensor pairing under coarse equivalence.

#### Novelty

- Judge: `NOVEL` over 0 arXiv hits

### Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_6db712c6.py", line 244, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_6db712c6.py", line 186, in run_trial
    kappa_f, kappa_g, kappa_fg = run_composition(f, g, k_f, k_g)
                                 ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_6db712c6.py", line 143, in run_composit
    X_f, d_f = build_X_d(f, k_f)
               ^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_6db712c6.py", line 43, in build_X_d
    X = [tuple(i) for i in range(n)]
         ^^^^^^^^
TypeError: 'int' object is not iterable
```

### Judge reasoning

The test crashed with a TypeError ('int' object is not iterable) at build_X_d before any κ values were computed, so no data exists to evaluate the pre-registered support criterion. | next: Fix build_X_d to enumerate vertices correctly (e.g., X = [tuple(b) for b in itertools.product([0,1], repeat=n)]) and rerun the 25-pair × 5-seed protocol.

---

## Coboundary-Vanishing Threshold: HX^1 Triviality Below the Diameter-to-Depth Ratio for Parity-on-a-Tree

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Coarse Geometric Karchmer-Wigderson (CG-KW) framework / coarse (Roe) cohomology and geometric group theory applied to circuit complexity × Formula depth lower bounds for the recursive-parity (balanced AND/OR-tree of XORs) function in the NC^1 vs. AC^0[⊕] regime, expressed via the coarse 1-cohomology HX^1(X_f) and the Roe-trace pairing on C_u^*[X_f]
- **Recorded**: 2026-04-28 03:39 UTC
- **Entry ID**: `264c40c0d040`

### Statement

Let f_n = ⊕-tree_n be the balanced binary tree of XOR gates on n = 2^k inputs, with KW-metric space (X_{f_n}, d_{f_n}). Define the coboundary δ : C^0_coarse(X_{f_n}) -> C^1_coarse(X_{f_n}) and the propagation-R Roe subalgebra C_u^*[X_{f_n}]_R. Then there exists a universal constant α ∈ (0,1) such that, for all R ≤ α·log_2 n, every coarse 1-cocycle c with propagation ≤ R is a coboundary (so its class in HX^1 is trivial), while for R ≥ k = log_2 n there exist nontrivial classes c whose Roe-trace pairing satisfies log|⟨c,T⟩| / log(propagation(T)) ≥ k − O(1). Consequently κ(f_n) ≥ log_2 n − O(1), matching depth(f_n) = Θ(log n) up to constants and certifying axiom A1 sharply for an explicit recursive function.

### Rationale

This conjecture directly tests axiom A1 (Coarse-KW link) together with A5 (Roe-index rigidity) on the canonical recursive function whose formula depth is exactly known (depth(⊕-tree_n) = log_2 n).

The KW protocol for the parity tree must descend the tree to find a differing leaf, so the controlled-cover multiplicity must grow by 1 per level; this should manifest as a vanishing-threshold for HX^1 at propagation R ≈ log_2 n, with cocycles at smaller propagation forced to be coboundaries. Verifying both the vanishing below the threshold and the nonvanishing above gives a sharp quantitative version of A1 on a function whose depth we can independently certify, providing a sanity check that κ is neither overshooting nor undershooting the true formula depth in a controlled recursive setting.

### Novelty

- Judge: `NOVEL` over 9 arXiv hits

Top hits consulted: - [2107.05128v2] Karchmer-Wigderson Games for Hazard-free Computation - [2002.07444v1] Multiparty Karchmer-Wigderson Games and Threshold Circuits - [2109.13292v1] The Hochschild Cohomology of Uniform Roe Algebras - [1904.05483v2] Parallels Between Phase Transitions and Circuit Complexity? - [1906.11725v2] On the uniform Roe algebra as a Banach algebra and embeddings of $\ell_p$ uniform Roe algebras

### Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_a7e3dd4e.py", line 133, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_a7e3dd4e.py", line 73, in run_trial
    tree = xor_tree(n)
           ^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_a7e3dd4e.py", line 20, in xor_tree
    left = xor_tree(n // 2)
           ^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_a7e3dd4e.py", line 22, in xor_tree
    return [left[i] ^ right[i] for i in range(n)]
            ~~~~^^^
IndexError: list index out of range
```

### Judge reasoning

The test crashed with an IndexError in the xor_tree construction before any data on R*(n), dim HX^1_R, or slope correlations was produced, so neither the support nor falsification conditions can be evaluated. | next: Fix the xor_tree recursion (the base case and/or input length handling causing the out-of-range index) and rerun the pre-registered protocol across n ∈ {4,8,16,32}.

---

## Multiplicity-Doubling Lower Bound from Protocol Pullback on Indexing Gadgets

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Coarse Geometric Lifting (CGL) / Coarse Geometry & Roe-Algebra methods in Communication Complexity × Deterministic two-party communication complexity CC(f∘G^n) of lifted boolean functions
- **Recorded**: 2026-04-28 08:37 UTC
- **Entry ID**: `6a6a970e1f5b`

### Statement

Let $G = IND_b$ be the standard indexing MetricGadget on $X = \{0,1\}^b \times [b]$, $Y = [b]$ with d the Hamming metric on $X\times Y$, and let $f: \{0,1\}^n \to \{0,1\}$ have deterministic decision-tree complexity $Q(f)$. Then for the lifted input space $((X\times Y)^n, d_\oplus)$ at scale $R = diam(G)$, every deterministic protocol $\Pi$ computing $f\circ G^n$ satisfies $CoarsePullbackProtocol(\Pi) = (m_\Pi, R_\Pi)$ with $m_\Pi \geq 2^{Q(f)} / poly(b, n)$, so by axiom A2, $CC(f\circ G^n) \geq Q(f) - O(\log(bn))$. Equivalently, $CLC(f, IND_b) \geq Q(f) \cdot \log_2(asdim_R(IND_b^{\otimes n})+1) - O(\log bn)$, recovering the Raz–McKenzie bound through the coarse-cover route.

### Rationale

This sub-conjecture directly tests axiom A2 (Protocol → Cover) on the canonical gadget for which Raz–McKenzie lifting is known to hold, using axiom A3 (Asdim Amplification) to guarantee that the iterated indexing gadget has unbounded coarse dimension proportional to n. If A2 is sound, the multiplicity of the protocol-induced RoeCover must scale exponentially in the query complexity, otherwise one could collapse a query-tree lower bound through a low-multiplicity cover — contradicting the structure theorem at the heart of CGL. Confirming the bound on small parameters provides empirical evidence that the coarse-cover viewpoint reproduces the standard query-to-communication lift, a necessary sanity check before pursuing axiom A4's non-relativizing strengthening.

### Novelty

- Judge: `NOVEL` over 9 arXiv hits

Top hits consulted: - [2211.17214v1] Lifting to Parity Decision Trees Via Stifling - [2105.01963v3] One-way communication complexity and non-adaptive decision trees - [2512.11833v1] Soft Decision Tree classifier: explainable and extendable PyTorch implementation - [1904.13056v4] Query-to-Communication Lifting Using Low-Discrepancy Gadgets - [2012.03415v1] On Query-to-Communication Lifting for Adversary Bounds

### Empirical Test

- exit code: `1`, elapsed: `0.11s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_38168056.py", line 114, in <module>
    seeds = [int(x) if x else 11 for x in input().split()]
                                          ^^^^^^^
EOFError: EOF when reading a line
```

### Judge reasoning

The test crashed with an EOFError while reading seeds from stdin before any data could be produced, so neither the support nor falsification conditions can be evaluated. | next: Patch the harness to provide default seeds (or read from argv/env) when stdin is empty, then rerun the ≥20-trial battery over $b\in\{2,3,4\}$, $n\in\{2,3,4\}$.

---

## $\Sigma^b_0$-PIND Width-2 Closure Round-Count vs DPLL Leaves

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Bounded arithmetic / Cook-Reckhow-Krajicek correspondence: the $\Sigma^b_0$-PIND round count $r_2(F)$, defined as the minimum $k \in \mathbb{N} \cup \{\infty\}$ such that iterating the closure operator $\Phi(S) = S \cup \{C : |C| \leq 2$ and C is RUP-derivable from $F \cup S$ in one unit-propagation pass$\}$ k times yields $\bot \in \Phi^k(\square)$; $r_2$ captures the depth of sharply-bounded polynomial-time

induction in PV restricted to width-2 reasoning (Cook-Reckhow lineage; rarely measured directly as a quantitative invariant on small unsatisfiable CNFs). × Lex-DPLL search-tree leaf count L(F) on small unsatisfiable 3-CNF formulas under a fixed lexicographic variable order with unit propagation, equivalently tree-resolution leaf count up to constants.
- **Recorded**: 2026-04-28 08:46 UTC
- **Entry ID**: `fc9124ecdaa8`

### Statement

For every unsatisfiable 3-CNF F on $n \leq 14$ variables and $m \leq 6n$ clauses: (a) if $r\_2(F) < \infty$, then $\log_2 L(F) \leq 2 \cdot r\_2(F) + 2 \cdot \log_2(n+1)$; (b) if $r\_2(F) = \infty$, then $\log_2 L(F) \geq n/4$. Equivalently, finiteness of width-2 RUP closure forces a polynomial DPLL bound, while non-closure forces an exponential one — a sharp dichotomy.

### Rationale

Width-2 RUP closure is the propositional shadow of Cook's PV restricted to the sharply-bounded Σ^b_0-PIND fragment, the weakest layer of the Buss hierarchy. The Cook-Reckhow-Krajicek correspondence predicts a polynomial relation between sharply-bounded PV-proof depth and tree-like resolution size; quantifying it should expose the exact slack and yield a dichotomy mirroring the classical PHP/Tseitin obstructions to weak proof systems.

### Novelty

- Judge: `NOVEL` over 11 arXiv hits

Top hits consulted: - [0209032v3] Complexity Results on DPLL and Resolution - [2407.17947v2] Supercritical Size-Width Tree-Like Resolution Trade-Offs for Graph Isomorphism - [1502.02131v2] Extracting verified decision procedures: DPLL and Resolution - [1411.7087v6] Consistency proof of a fragment of PV with substitution in bounded arithmetic - [2306.08535v1] Cyclic proofs for arithmetical inductive definitions

### Empirical Test

- exit code: `124`, elapsed: `180.07s`

```
TIMEOUT after 180s
```

### Judge reasoning

The test timed out after 180s (returncode=124) without producing any data, so neither the support nor falsification conditions could be evaluated. The critic also challenges, and no instances were verified. | next: Reduce the search space (e.g., cap n at 10 and reduce seeds per (n,α) pair) or parallelize/optimize the r_2 closure and DPLL leaf-count routines so the test completes within the time budget.

---

## Mahler Measure of Truth-Table Polynomial Lower-Bounds DNF_min

- **Verdict**: `BARRIER_HIT`
- **Bridge**: Number theory of polynomial heights — the (logarithmic) Mahler measure m(p) = log|a_d| + sum_{|α|>1} log|α| over the complex roots of p ∈ Z[x] (Lehmer-Boyd-Smyth tradition: Lehmer's problem on smallest non-cyclotomic measure, Boyd's heights, Smyth's resolved cubic case). Rarely deployed in meta-complexity; computable in poly time via numerical root-finding on the explicit degree-(2^n − 1) integer polynomial. × DNF_min(f): the minimum number of product terms in a DNF representation of a Boolean function f:{0,1}^n→{0,1} given its

2^n-bit truth table — the Hirahara–Oliveira–Santhanam restricted-MCSP proxy at the heart of meta-complexity worst-case-to-average-case reductions.
- **Recorded**: 2026-04-28 09:18 UTC
- **Entry ID**: `a6aec390c941`

### Statement

Define the truth-table polynomial p_f(x) := sum_{i=0}$^{\{2}$n − 1} f(i)·x^i ∈ Z[x], where f(i) is the value of f on the n-bit binary expansion of i. For every f with |f^{-1}(1)| ≥ 1, m(p_f) ≤ (DNF_min(f) − 1) · log_2(n + 1). Equivalently, every f whose truth-table polynomial has logarithmic Mahler measure exceeding (k − 1)·log_2(n+1) satisfies DNF_min(f) > k.

### Rationale

Each DNF product term corresponds to a coordinate-aligned subcube whose index set is a coset a+W ≤ F_2^n; its indicator polynomial factors as x^a · prod_j (1 + x$^{\{2}${i_j}}), so all roots lie on the unit circle and m = 0 (purely cyclotomic). Disjunction of k subcubes makes p_f a signed sum of k such cyclotomic-rooted polynomials, and Boyd-style sub-additive heuristics for Mahler measure under integer-coefficient sums predict at most a log_2(deg+1) escape per added term. If true, this transforms a hard Diophantine height into a poly-time-computable lower bound on DNF_min, giving a Lehmer-type obstruction useful in restricted-MCSP regimes.

### Novelty

- Judge: “ over 0 arXiv hits

### Empirical Test

- exit code: `-1`, elapsed: `0.00s`

```
(no output)
```

### Judge reasoning

Rejected by NATURAL_PROOFS barrier

---

## Plünnecke-Ruzsa Doubling of Constraint Vectors Lower-Bounds SoS Degree for 3-XOR

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Additive combinatorics in F_2^n: the Plünnecke-Ruzsa doubling constant K(A)=|A+A|/|A| and its log version σ(A)=log_2|A+A|-log_2|A| of a finite set A⊆F_2^n (Freiman-Ruzsa / Bogolyubov / Sanders’ polynomial Bogolyubov tradition). The mathematical branch is harmonic / arithmetic combinatorics on abelian groups, distinct from polyhedral / Newton-polytope geometry, and rarely deployed in SoS lower bounds for CSPs. × Sum-of-Squares refutation degree d_SoS(F) of unsatisfiable 3-XOR instances F over GF(2), accessed via the Grigoriev-Schoenebeck combinatorial proxy g*(F) = min{|T| : ⊕*{i∈T} a_i = 0 ∈ F_2^n and* ⊕{i∈T} b_i = 1 ∈ F_2}, which sandwiches SoS degree to within ±1 by the Schoenebeck (2008) characterization for 3-XOR.
- **Recorded**: 2026-04-28 09:45 UTC
- **Entry ID**: `6ae7590ec4aa`

**Statement**

For every unsatisfiable 3-XOR instance $F = ((a_i,b_i))_{i=1..m}$ on $n \leq 20$ variables with constraint multiset $A = \{a_1,...,a_m\} \subseteq F_2^n$, the Schoenebeck combinatorial SoS-degree satisfies $g(F) \geq \sigma(A) + 1$, where $\sigma(A) = log_2|A+A| - log_2|set(A)|$. Equivalently, $|A+A| \leq |set(A)| \cdot 2^{g(F)-1}$. A single (F, A) violating this inequality refutes the conjecture.

**Rationale**

Schoenebeck's theorem says small SoS degree forces a short additive dependency among constraint vectors; Plünnecke-Ruzsa says short additive dependencies push A toward an affine-subspace structure with low doubling. The bridge predicts a monotone, never-before-quantified link between structurelessness (large doubling) of the constraint vectors and the unavoidable length of a Schoenebeck certificate, transferring Freiman-Ruzsa intuition into proof complexity.

**Novelty**

- Judge: `NOVEL` over 9 arXiv hits

Top hits consulted: - [2507.01005v1] Discovery and study of the eclipsing variable star Grigoriev 1 - [0701009v2] Electron mobility on a surface of dielectric media: influence of surface level atoms - [1812.10216v2] Robust Approach for Rotor Mapping in Cardiac Tissue - [1701.04521v1] Sum of squares lower bounds for refuting any CSP - [0305179v3] Polynomial Degree and Lower Bounds in Quantum Complexity: Collision and Element Distinctness with Small Range

**Empirical Test**

- exit code: `1`, elapsed: `0.11s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_16090c09.py", line 92, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_16090c09.py", line 70, in run_trial
    set_A_size, A_plus_A_size, sigma_value = sigma(A)
                                             ^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_16090c09.py", line 56, in sigma
    A_plus_A.add(xor(a, b))
                 ^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_16090c09.py", line 21, in xor
    return x ^ y
           ~~^~~
TypeError: unsupported operand type(s) for ^: 'list' and 'list'
```

**Judge reasoning**

The test crashed with a TypeError in the xor() helper (xoring two lists), so no data was produced and the pre-registered support condition could not be evaluated. | next: Fix the xor() helper to operate on integer bitmask representations of F_2^n vectors (or convert lists via tuple/int) and re-run the 3000-instance sweep.

---

## Kolmogorov Width of Lifted Functions Bounds MA^cc

- **Verdict**: `INCONCLUSIVE`

- **Bridge**: Descriptive complexity theory via Kolmogorov complexity × MA communication complexity
- **Recorded**: 2026-04-28 10:22 UTC
- **Entry ID**: `dd203ee122c9`

### Statement

For any Boolean function f, the Kolmogorov width of the lifted function f ∘ G^n is at least Ω(log(n)) times the MA communication complexity of f, where G^n is the n-bit Indexing gadget.

### Rationale

Kolmogorov complexity and descriptive complexity theory provide a framework for understanding the intrinsic complexity of objects, which can be applied to communication complexity to derive lower bounds. The use of lifting gadgets, such as the Indexing gadget, allows us to amplify the complexity of the function, making it more amenable to analysis.

### Novelty

- Judge: `NOVEL` over 8 arXiv hits

Top hits consulted: - [1811.04010v2] The layer complexity of Arthur-Merlin-like communication - [1904.13056v4] Query-to-Communication Lifting Using Low-Discrepancy Gadgets - [0506068v1] Quantum Arthur-Merlin Games - [1409.0584v2] Kolmogorov structure functions for automatic complexity - [1101.2946v3] Quantum Kolmogorov Complexity and Information-Disturbance Theorem

### Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_9461c469.py", line 78, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_9461c469.py", line 43, in run_trial
    lifted_f = [f[(i >> n) * (1 << n) + i % (1 << n)] for i in range(2**(n+n))]
                ~^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
IndexError: list index out of range
```

### Judge reasoning

The test crashed with an IndexError before producing any data, so neither the support nor falsification conditions could be evaluated. No empirical evidence exists to assess the conjecture. | next: Fix the indexing bug in the lifted function construction (the formula `(i >> n) * (1 << n) + i % (1 << n)` is malformed for the Indexing gadget) and rerun the 120 trials.

---

## Tensor-Amplified Asdim Forces Linear Communication Blow-up on Inner-Product Gadgets

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Coarse Geometric Lifting (CGL) / Coarse Geometry & Asymptotic Dimension (geometric group theory applied to communication complexity) × Deterministic two-party communication complexity of lifted functions f∘G^n, with G obtained by k-fold GadgetComposition of a base inner-product gadget IP_2

- **Recorded**: 2026-04-28 10:39 UTC
- **Entry ID**: `916caa07c776`

### Statement

Let IP_2 be the 2-bit inner-product MetricGadget on X=Y={0,1}^2 with Hamming metric d, and let G_k = IP_2^{⊗k} under coordinatewise additive metric. For every Boolean f:{0,1}^n→{0,1} with deterministic query complexity Q(f), the CoarseLiftingComplexity satisfies CLC(f, G_k) ≥ Q(f) · log_2(k+1) − O(n), and consequently CC(f∘G_k^n) ≥ Q(f)·log_2(k+1) − O(n). Equivalently, AsdimCertify(LiftedInputSpace(G_k), R=diam(G_k)) returns multiplicity m ≥ k+1, so by axiom A2 every deterministic protocol Π must have cost ≥ log_2(k+1) per query bit recovered.

### Rationale

This sub-conjecture directly tests axiom A3 (Asdim Amplification) chained with A2 (Protocol→Cover). A3 predicts that iteratively tensoring IP_2 (which already has asdim ≥ 1 because its 16-point input space contains a metric line of length 4) yields G_k with asdim ≥ k − O(1). A2 then converts the cover multiplicity into a communication lower bound. If the conjecture holds, it provides a concrete constructive route to lifting bounds that scale with the tensor depth k, recovering Raz–McKenzie–style log-factor blow-ups from a purely coarse-geometric witness rather than from gadget-specific combinatorics. Failure would falsify A3 in the small-multiplicity regime and force a refinement of how asdim composes under ⊗.

### Novelty

- Judge: `NOVEL` over 9 arXiv hits

Top hits consulted: - [1904.13056v4] Query-to-Communication Lifting Using Low-Discrepancy Gadgets - [1104.4468v3] A strong direct product theorem for quantum query complexity - [2511.17227v3] A Lifting Theorem for Hybrid Classical-Quantum Communication Complexity - [0111062v2] One-way communication complexity and the Neciporuk lower bound on formula size - [1110.6876v4] Lower bounds for topological complexity

### Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_af1c65ed.py", line 185, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_af1c65ed.py", line 124, in run_trial
    G_k = gadget_composition([ip_2] * k)
          ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_af1c65ed.py", line 74, in gadget_compos
    result[i*n+k][j*n+l] = gadgets[i][k] + gadgets[j][l]
                           ~~~~~~~~~~^^^
TypeError: 'function' object is not subscriptable
```

### Judge reasoning

The test crashed with a TypeError in gadget_composition (treating a function as subscriptable) before any data was produced, so neither the support nor falsification criteria could be evaluated. | next: Fix the gadget_composition implementation to operate on metric matrices/arrays rather than function objects, then re-run the 12 (k,n,f) instances across 3 seeds.

---

## Möbius Defect of NW Design Lattice Bounds Parity Bias

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Möbius theory of finite lattices (Rota–Stanley combinatorial Möbius inversion on intersection lattices; Stanley EC1 Ch.3) — rarely deployed in PRG bias analysis × Output bias of the Nisan–Wigderson pseudorandom generator NW_D$^{f:\{0,1\}}$d→{0,1}^m against linear (parity) distinguishers χ_T, T⊆[m]
- **Recorded**: 2026-04-28 10:48 UTC
- **Entry ID**: `b27f028c321c`

### Statement

Let D=(S_1,…,S_m) be an NW design over universe [d] with |S_i|=n and pairwise intersections ≤k. Let L(D) be the meet-semilattice of all distinct intersections T_I=⋂*{i∈I} S_i (I⊆[m]) ordered by ⊆ with bottom 0̂=∅, and let μ_L be its Möbius function. Define the Möbius defect δ(D)=Σ*{T∈L(D), T≠∅} |μ_L(0̂,T)|·2^{−|T|}. Conjecture: for every f:{0,1}^n→{0,1} with max Walsh coefficient |f̂(U)|≤2^{−n/3} on |U|≥k, and every non-empty T⊆[m], |E_{y∼U_d}[χ_T(NW_D^f(y))]| ≤ 4·δ(D).

### Rationale

Standard NW bias bounds use the crude factor m·2·ε from a hybrid argument that ignores higher-order inclusion–exclusion cancellation across the design's intersection pattern. Möbius values μ_L(0̂,T) are precisely the signed multiplicities by which intersections of design rows are over/under-counted, so replacing m by δ(D) should yield substantially tighter bias estimates whenever L(D) is structured (e.g., projective-plane or polynomial designs versus random subset-systems). If the conjecture holds, δ(D) becomes a computable, design-only certificate of fooling power independent of f beyond the Fourier-tail hypothesis.

### Novelty

- Judge: `NOVEL` over 9 arXiv hits

Top hits consulted: - [2601.03730v1] Perception-Aware Bias Detection for Query Suggestions - [1207.0393v2] Pseudo-finite hard instances for a student-teacher game with a Nisan-Wigderson generator - [2109.14047v1] Second Order WinoBias (SoWinoBias) Test Set for Latent Gender Bias Detection in Coreference Resolution - [0710.4339v1] Heavy-Quark Masses from the Fermilab Method in Three-Flavor Lattice QCD - [1111.0981v1] Form factors for $B$ to $Kll$ semileptonic decay from three-flavor lattice QCD

### Empirical Test

- exit code: `0`, elapsed: `0.03s`


```
{"TRIAL": null}
{"TRIAL": null}
{"TRIAL": null}
{"TRIAL": null}
{"TRIAL": null}
```


### Judge reasoning

The test stdout contains only `{"TRIAL": null}` entries with no measurements, no RESULT line, and no evidence that the pre-registered criteria (≥200 designs, both constructions, regression slope ≥0.5, zero violations) were evaluated. The critic also flags a likely vacuous-bound saturation issue, so SUPPORTED cannot be claimed. | next: Fix the test harness so trials actually emit (D,f,T,bias,δ(D)) tuples, then verify whether 4·δ(D) is non-trivial (<1) for the tested regime before re-running the p

## Sandpile-Group Order of Certificate Conflict Graph Lower-Bounds Q^dt

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Abelian sandpile models / chip-firing on finite graphs (Bjorner–Lovász–Shor; Holroyd–Levine–Mészáros–Peres–Propp–Wilson); the sandpile (critical) group K(G) computed as the cokernel of the reduced graph Laplacian, equivalently the spanning-tree count by Kirchhoff's matrix-tree theorem × Decision-tree depth Q^dt(f) of small Boolean functions f:{0,1}^k→{0,1}, transported via the Raz–McKenzie / Göös–Pitassi–Watson Indexing-gadget lifting theorem to deterministic two-party communication complexity D^cc(f∘IND_b)
- **Recorded**: 2026-04-28 11:19 UTC
- **Entry ID**: `ef886a7094f0`

### Statement

For every Boolean f:{0,1}^k→{0,1}, let G_f be the graph whose vertices are the minimal 1-certificates of f and whose edges {C,C'} connect certificates that fix some common variable to opposite values (call these 'conflicts'). Then log_2 |K(G_f)| ≤ Q^dt(f) · log_2(k+1), where |K(G_f)| is the order of the abelian sandpile group of G_f, which equals the number of spanning trees of G_f by Kirchhoff (defined to be 1 if G_f has ≤1 vertex or is disconnected with each component a tree). By GPW lifting, this entails D^cc(f∘IND_b) ≥ Ω(log_2 |K(G_f)| · log b / log k).

### Rationale

The minimal 1-certificate clutter captures the irreducible branching obligations any decision tree must witness; their conflict graph encodes exactly the pairs that cannot coexist on a single root-to-leaf path, so its cycle-richness should constrain protocol/decision-tree branching budgets. The sandpile group is a torsion abelian invariant whose log-size is Kirchhoff's spanning-tree count — a single integer determinant — and a depth-d tree induces at most k^d distinct query histories, plausibly bounding how much such cyclic redundancy can be 'covered'. Chip-firing and sandpile invariants have essentially never been deployed in lifting-style lower bounds, giving an algebraic-combinatorial obstruction distinct from rank, Dilworth-width, Möbius, or asymptotic-dimension routes.

### Novelty

- Judge: `NOVEL` over 9 arXiv hits

Top hits consulted: - [0906.2809v2] Sandpile groups and spanning trees of directed line graphs - [2602.01042v2] On Condensation of Block Sensitivity, Certificate Complexity and the `AND` (and `OR`) Decision Tree Co - [1904.12209v2] Sandpile monomorphisms and limits - [2512.11833v1] Soft Decision Tree classifier: explainable and extendable PyTorch implementation - [0811.2800v2] Parallel Chip-Firing on the Complete Graph: Devil's Staircase and Poincare Rotation Number

### Empirical Test

- exit code: `1`, elapsed: `0.11s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_6f6dc07d.py", line 131, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^

  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_6f6dc07d.py", line 104, in run_trial
    if not is_minimal_1_certificate(f, C):
           ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
```

```
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_6f6dc07d.py", line 25, in is_minimal_1_
    if f[C[:i] + (1 - C[i],) + C[i+1:]] == f[C]:
         ~~~~~~^~~~~~~~~~~~~
TypeError: can only concatenate list (not "tuple") to list
```

**Judge reasoning**

The test crashed with a TypeError (list/tuple concatenation bug in is_minimal_1_certificate) before any of the 800 trials produced data, so the pre-registered support condition cannot be evaluated. | next: Fix the type bug by representing certificates consistently (e.g., always tuples) so that C[:i] + (1 - C[i],) + C[i+1:] type-checks, then rerun the 800-trial sweep.